

Title

WHONDRS Surface Water and Sediment Geochemistry and Organic Matter Characterization Data from Streams across HJ Andrews Experimental Forest, Oregon (v2)

Summary

This dataset supports a broader study developing conceptual models for river corridor critical zone processes across spatial scales and was generated in collaboration with the HJ Andrews River Corridor Critical Zone Workshop in 2025. The dataset provides surface water geochemistry (dissolved organic carbon, total dissolved nitrogen) from 48 sites across the HJ Andrews Experimental Forest, Oregon (<https://andrewsforest.oregonstate.edu>). Some of the sites have been impacted by the Holiday Farm Fire and the Lookout Fire in 2020 and 2023, respectively. Related data were collected as part of the workshop and will be published separately in collaboration with other workshop attendees and available at <http://www.hydroshare.org/resource/b274c4a234bf4b12b7cb8a54a696c629>. Related genomic data can be found on the National Center for Biotechnology Information (NCBI) under BioProject PRJNA1503030 (see critical details section below for more information).

Additional related data collected in 2016 from a similar effort can be found at <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/3377027> and <http://www.hydroshare.org/resource/ea6c0832885a46c3939e7bb22e48e754> and are described within <https://doi.org/10.5194/essd-11-1-2019> (Ward et al., 2019).

This data package was originally published in March 2026. It was updated in August 2026 (v2; new and modified files). See the change history section below for more details.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

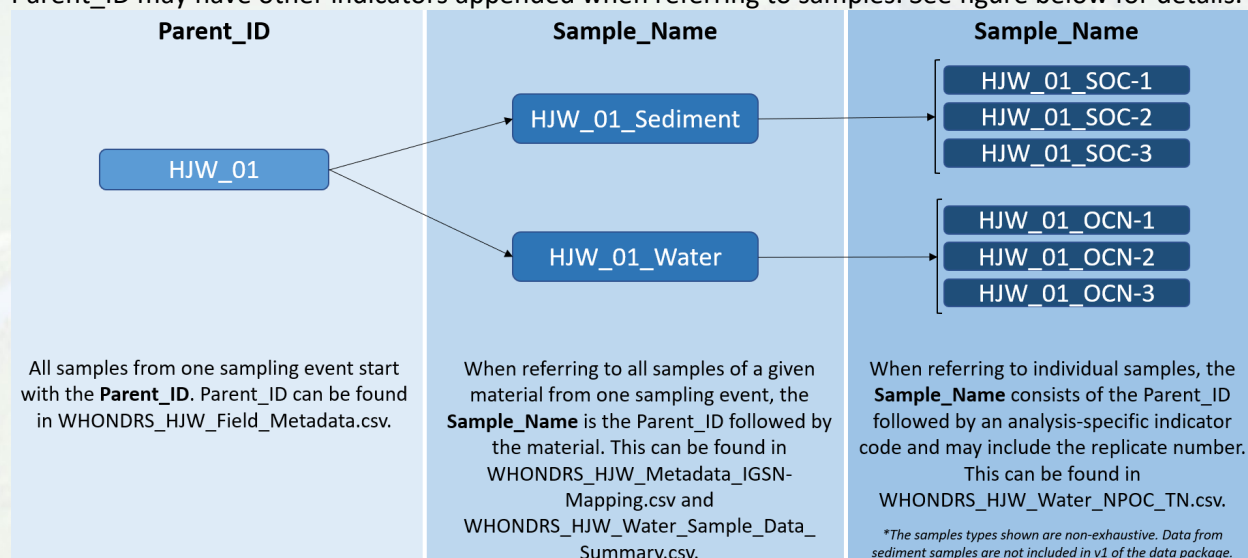
Brief Overview of Methods

Samples were collected on July 27 – August 1, 2025 from 48 sites across HJ Andrews Experimental Forest, Oregon, United States via WHONDRS (<https://whondrs.pnnl.gov>) and HJ Andrews River Corridor Critical Zone Workshop collaboration. Site metadata, including general environmental information, were manually recorded in the field. See WHONDRS_HJW_Field_Protocol.pdf for details on sample collection and site metadata collection. Briefly, surface water samples were collected at 50 percent water column depth and syringe-filtered. Sediment samples were collected underwater at 1 to 3 centimeters depth below the streambed and sieved to less than 2 millimeters. The water samples were refrigerated after sampling and shipped on blue ice to Pacific Northwest National Laboratory. The sediment samples were stored on dry ice upon collection and shipped with dry ice to Pacific Northwest National Laboratory. For details regarding laboratory and instrument methods for this analysis and others, see the alphanumeric methods codes located in the header rows of the chemistry sample data csv files and their associated definitions in WHONDRS_HJW_Methods_Codes.csv. Data formats follow ESS-DIVE Reporting Formats.

Critical Details

- 1 – **Sample naming:** Each sampling event has a unique Parent_ID in the format HJW_###. The field metadata and data files all contain these unique IDs and can be mapped across each other accordingly. The

Parent_ID may have other indicators appended when referring to samples. See figure below for details.



- 2 – **File naming:** The files within this data package follow a consistent naming scheme to enable easy use of the data and support AI-readiness. All files begin with “WHONDRS” and a three-letter code that indicates the study – “HJW”. The text following these indicators and separated with an underscore describes the material and data type(s) within the file.
- 3 – **File joining:** The file-level metadata (FLMD) file includes a column that lists the fields in each file to facilitate joining. When joining sample data with field metadata or other sample data, join based on the Parent ID portion of the Sample Name (i.e., the analyte code and replicate number should be ignored). Replicate numbers should not be used for joining. They do not represent paired samples across different analyses (i.e., replicate 1 from one data type is often a different sub-sample than replicate 1 of another data type). There is one exception to this - NPOC data is used for determining dilution for FTICR analysis and therefore replicate 1 in the NPOC data is the same as replicate 1 in the FTICR data.
- 4 – **FTICR-MS processing levels:** FTICR-MS data is provided in four processing levels in this data package. The unprocessed instrument files are available in a zipped folder called “WHONDRS_HJW_FTICR_Raw_Data.zip”. These can be processed in newly developed public workflows (i.e., CoreMS) for data use or they can be first processed in a proprietary software (i.e., Bruker) into XML files and then processed in publicly available Formultitude (previously called Formularity; <https://github.com/PNNL-Comp-Mass-Spec/Formultitude>) for further data use. In previous data packages from this project, we have processed the files in Bruker prior to publishing and then only published XML files and instructions for the user to process the XML files in Formultitude. In this data package, due to these new software developments, we have included raw data, pre-processed (i.e., peak picked) XML files, CoreMS output files, and fully processed data to allow for maximum data user flexibility.
- 5 – **FTICR-MS processing scripts:** We have provided a static version of our CoreMS processing instructions and scripts that were used for this dataset. These instructions and scripts may change over time as this new pipeline evolves. The most up to date instructions and scripts can be found at https://github.com/river-corridors-sfa/RCSFA_CoreMS. A release was created on GitHub to document the version pulled for this data package.

- 6 – **FTICR-MS merging across datasets:** We do not recommend combining **processed** FTICR-MS data, whether it was run through Formultitude or CoreMS (*CoreMS_Processed_ICR_*.csv). If you wish to combine data from across datasets, you need to go back to at least the .xml files and reprocess the data using your pipeline of choice. For convenience, we have included the CoreMS outputs (*.corems.csv files) which **can** be combined with other CoreMS outputs released in our data packages, though we still recommend reprocessing with your pipeline. We are happy to work through these considerations with you if you have questions.
- 7 – **FTICR Blank Correction:** We recommend blank subtraction when processing these data. The provided processed data is blank corrected. Any formula that was found in either of the two blanks was removed from the dataset.
- 8 – **Genome Data:** Related genomic data can be found on the National Center for Biotechnology Information (NCBI) under BioProject PRJNA1503030. Briefly, sequencing libraries were constructed by extracting nucleic acids with the Qiagen PowerSoil DNA Extraction kits, amplifying the V4 region of the 16S rRNA gene with the PCR, and pooling barcoded PCR products at equimolar ratios. Libraries were sequenced on an Illumina MiSeq at the Oregon State University Center for Quantitative Life Sciences, and the 2x300bp reads were archived on NCBI.

Data Package Structure

In addition to this readme, this data package also includes a file-level metadata (FLMD) file that describes each file and a data dictionary (DD) that describes all column/row headers and variable definitions.

This dataset is comprised of (1) a folder of field photos, (2) a folder of raw Fourier transform ion cyclotron resonance mass spectrometry (FTICR-MS) data, (3) a data checks report, (4) a folder of sample data, (5) file-level metadata, (6) data dictionary, (7) field metadata, (8) readme, (9) international generic sample number (IGSN) mapping file; and (10) field protocol. The sample data subfolder contains surface water and sediment (1) dissolved organic carbon (DOC, measured as non-purgeable organic carbon, NPOC) data and averages, (2) total dissolved nitrogen data and averages, (3) methods codes, (4) FTICR-MS methods; and (5) a subfolder of 9.4 Tesla (9.4T) FTICR-MS data. This folder contains the CoreMS processed data and seven subfolders, three containing .xml files for each sample type (sediment, surface water and blank samples), three containing the sediment CoreMS output files for each sample type (sediment, surface water and blank samples), and the other containing instructions and scripts for processing the files in CoreMS (<https://github.com/EMSL-Computing/CoreMS>). All files are .csv, .pdf, .R, .xml, .Rmd, .py, .cal, .json, .jpg, or .jpeg.

Citations, Acknowledgements, and License

Acknowledgments

This research was supported by the Worldwide Hydrobiogeochemistry Observation Network for Dynamic River Systems (WHONDRS) through the U.S. Department of Energy (DOE) Office of Science, Biological and Environmental Research (BER) Program, Environmental System Science (ESS) Program (<https://ess.science.energy.gov/>) through the Pacific Northwest National Laboratory (PNNL) River Corridor Science Focus Area (SFA). PNNL is operated by Battelle Memorial Institute for the DOE under Contract No. DE-AC05-76RL01830. Data and facilities were provided by the H.J. Andrews Experimental Forest and Long Term Ecological Research (LTER) program, administered cooperatively by Oregon State University, the USDA Forest Service Pacific Northwest Research Station, and the Willamette National Forest. This material is based upon work supported by the National Science Foundation under the grant

ILTER8 DEB-2025755. This research was also supported by NSF EAR-1904527, a Critical Zone Research Coordination Network. This work was supported in part by the U.S. Department of Energy, Office of Science, Biological and Environmental Research, Environmental System Science Program through the Watershed Dynamics and Evolution (WaDE) Science Focus Area at Oak Ridge National Laboratory (ORNL); ORNL is managed by UT-Battelle LLC for the U.S. Department of Energy under contract DE-AC05-00OR22725. FTICR-MS data were generated at University of Arizona by Malak Tfaily's lab (<https://malaktfaily.com>).

Citations:

- Ward, A. S., Zarnetske, J. P., Baranov, V., Blaen, P. J., Brekenfeld, N., Chu, R., Derelle, R., Drummond, J., Fleckenstein, J., Garayburu-Caruso, V., Graham, E., Hannah, D., Harman, C., Hixson, J., Knapp, J. L. A., Krause, S., Kurz, M. J., Lewandowski, J., Li, A., Martí, E., Miller, M., Milner, A. M., Neil, K., Orsini, L., Packman, A. I., Plont, S., Renteria, L., Roche, K., Royer, T., Schmadel, N. M., Segura, C., Stegen, J., Toyoda, J., Wells, J., Wisnoski, N. I., and Wondzell, S. M.: Co-located contemporaneous mapping of morphological, hydrological, chemical, and biological conditions in a 5th order mountain stream network, Oregon, USA, Earth Syst. Sci. Data. <https://doi.org/10.5194/essd-11-1-2019>
- Ward, A. (2024). River Corridor - ESSD - Data Collection (2024 Update), HydroShare, <http://www.hydroshare.org/resource/ea6c0832885a46c3939e7bb22e48e754>
- Ward, A. S. (2026). River Corridor Post-Fire Synoptic - ESSD, HydroShare, <http://www.hydroshare.org/resource/b274c4a234bf4b12b7cb8a54a696c629>
- Stegen J C ; Chu R ; Danczak R E ; Forbes B ; Garayburu-Caruso V A ; Goldman A E ; Graham E B ; Hager J R ; Renteria L ; Toyoda J G ; Synoptic Sampling Team T (2026): WHONDRS 2016 Sediment Organic Matter Characterization Data from Streams across HJ Andrews Experimental Forest, Oregon. River Corridor Hydro-biogeochemistry from Molecular to Multi-Basin Scales SFA, ESS-DIVE repository. Dataset. [doi:10.15485/3377027](https://doi.org/10.15485/3377027)

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Change History

Approach to change history and versioning:

Updates to **data package** version: When any file within a data package is updated, the data package version number is updated. The data package version number is indicated in the title of the data package, the data package folder name, and in the change history table below. You can access previous versions of the data package by sending a request to ESS-DIVE.

Updates to **individual file** versions: As files are changed, the file version number is also updated. The file version number is indicated in the file name, file level metadata (flmd) file, and the change history table below. The version number on an individual file may not match the version number of the data package. For example, v3 of a data package may include v2 of an individual file.

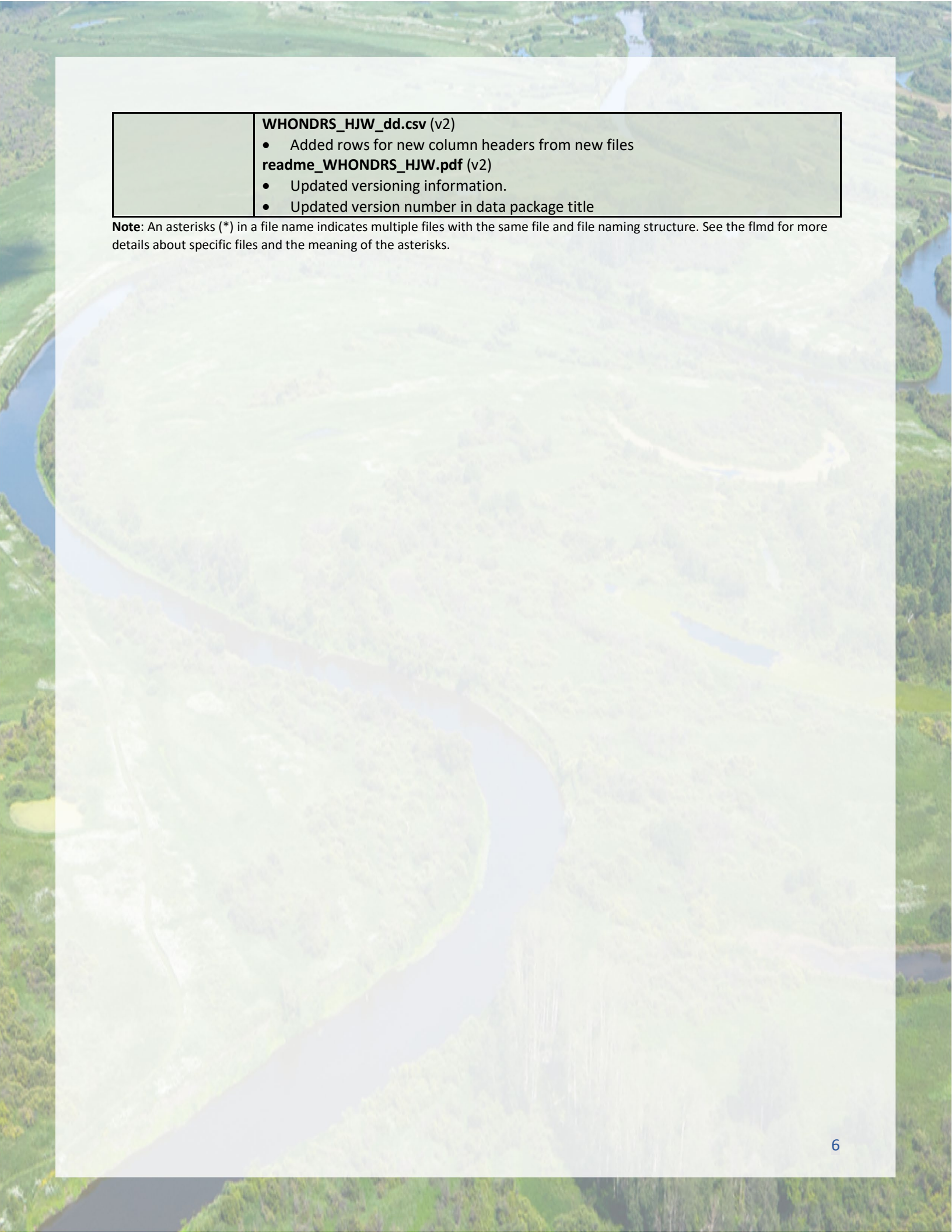
The change history below describes each file revised during versioning. If you are interested in seeing the exact cells within a file that have changed, you can utilize the daff package in R (<https://github.com/edwindj/daff>) to compare a previously downloaded file to a newly downloaded file.

In the change history table below, the sub-headers and bullets indicate the type of change in each file:

- **New files:** Describes new files added that were not present in previous data package versions.
- **Bulk changes to files:** Describes a change to many files within the data package. The indicated superscript will be added to each file name that the change applies to.
- **Modified files:**
 - **Corrected:** Describes existing information modified or removed to prevent sharing of incorrect information
 - **Added:** Describes new information inserted into an existing file (e.g., appending new columns/rows)
 - **Updated:** Describes modifying existing information to maintain accuracy though version changes. (e.g., changing version number to new version number)

Change history:

Data Package Version	Changes
Version 1 <i>March 2026</i>	Original data package publication
Version 2 <i>August 2026</i>	<u>NEW FILES</u> • WHONDRS_HJW_Sediment_NPOC_TN.csv • WHONDRS_HJW_Sediment_FTICR_Methods.csv • WHONDRS_HJW_Sediment_Sample_Data_Summary.csv • WHONDRS_HJW_*.xml • WHONDRS_HJW_*.corems.csv • WHONDRS_HJW_*.corems.json • WHONDRS_HJW_*.corems.cal • WHONDRS_HJW_*_CoreMS_Processed_ICR_Calibration.csv • WHONDRS_HJW_*_CoreMS_Processed_ICR_Data.csv • WHONDRS_HJW_CoreMS_Processed_ICR_Mol.csv • CoreMS_MergeProcess.Rmd • CoreMS_Runner.py • Format_CoreMS_Outputs.R • getLambda.R • Instructions_for_Processing_FTICR_Data_using_CoreMS.pdf • v2_WHONDRS_HJW_Data_Checks_Report.html <u>MODIFIED FILES</u> WHONDRS_HJW_Methods_Codes.csv (v2) • Added rows for new data types WHONDRS_HJW_flmd.csv (v2) • Added rows for new files • Updated version number for modified files.



	WHONDRS_HJW_dd.csv (v2) • Added rows for new column headers from new files readme_WHONDRS_HJW.pdf (v2) • Updated versioning information. • Updated version number in data package title
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Note: An asterisks (*) in a file name indicates multiple files with the same file and file naming structure. See the flmd for more details about specific files and the meaning of the asterisks.